

OncoFold

Visualizing cancer-associated mutations in
3D protein structures with *OncoFold*

Webinar | oncofold.org

Dietlein Lab • Boston Children's Hospital • Harvard Medical School

Workshop Overview

Part 1: Background

- The challenge of cancer genomics
- Proteo-genomics framework (HMM)
- Key findings across 173 genes

~15 min

Part 2: OncoFold Web Tool

- Interactive Visualizations
- Mutation Lookup
- Data Download & Resources

~30 min (incl. live demo & Q&A)

No prior bioinformatics or genomics expertise is required. The tool is built to be used by individuals from all backgrounds: clinicians, experimental biologists, computational researchers, and students alike.

The Challenge

Cancer Genomes

- Hundreds of mutated genes per tumor
- Roles in tumor development often remain incompletely understood
- Same gene, different domains → different effects
- Binary "driver vs. passenger" labels lack nuance for precision medicine

The Opportunity

- Population-scale sequencing: 139,818 patients
- 25+ tumor types analyzed across multiple cohorts
- AlphaFold predicts 3D structures for nearly all cancer genes
- Integrating protein structures with mutation data reveals spatial patterns

Who Is This For?

Clinical & Experimental

- Variant interpretation in patient genomes
- Connecting mutations to mechanistic hypotheses

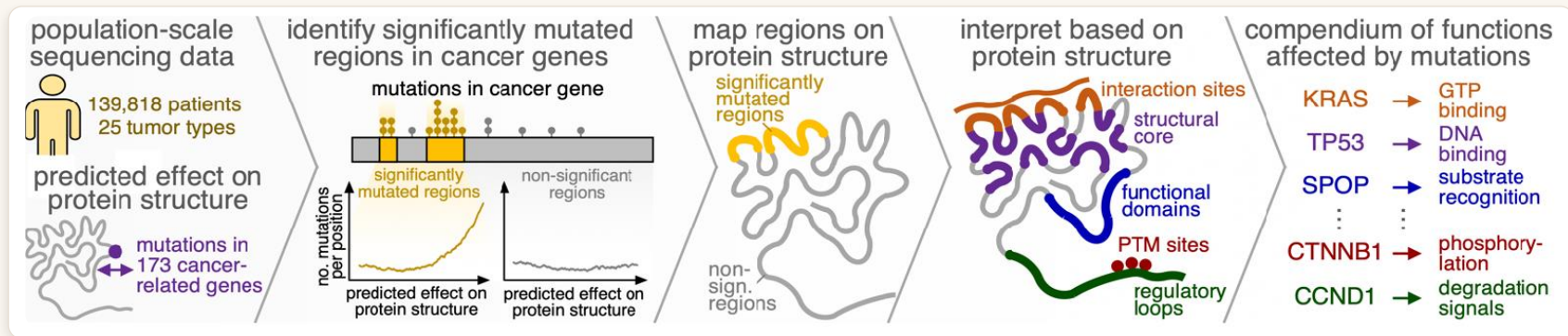
Computational & Precision Medicine

- From gene-level statistics toward subregion resolution
- Guiding therapeutic strategies in precision medicine

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The Proteo-Genomics Pipeline

From population-scale sequencing to a compendium of functions affected by mutations in cancer genes.



The Proteo-Genomics Framework

A hidden Markov model identifies "significantly mutated regions" — regions where mutations concentrate beyond what background rates predict.

1 Input Data

Data from 139,818 patients across >25 tumor types, harmonized and quality-filtered

2 Model

HMM accounts for underlying mutational patterns and constraints.

3 Output

Significantly mutated regions mapped to functional domains and 3D protein structures

Key advantage: directly interpretable at single base pair resolution, without requiring prior domain knowledge.

Key Findings

173

cancer genes
mapped

139,818

tumor patients
analyzed

25+

tumor types
covered

~22%

avg. gene length
significant

Functional domain concentration

DNA-binding (FOXA1), ligand-binding (ERBB2), catalytic (EP300), substrate-recognition (SPOP), degradation signals (CCND1)

Structural constraints

TP53: interior core. FBXW7: substrate sites. KRAS: GTP-proximal hotspots. 3D structures essential for full interpretation

Missense vs. truncating differ

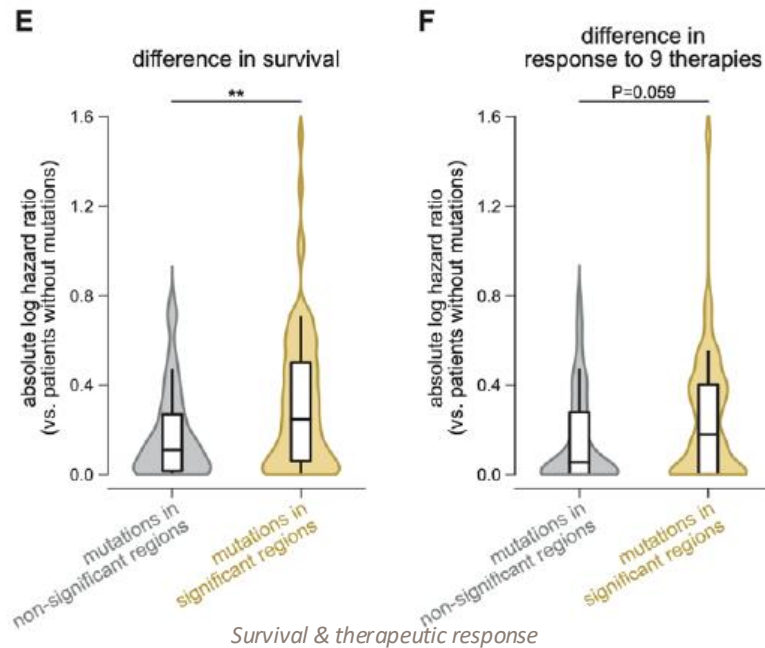
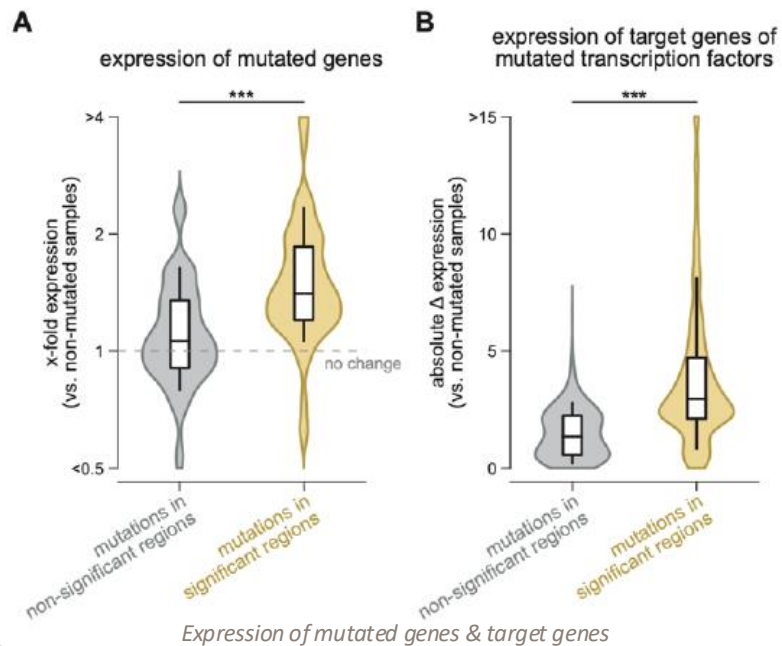
Missense: domain-specific. Truncating: broader but regionally biased — degradation signals in oncogenes, stable regions in tumor suppressors

Pan-cancer Findings

Significant regions mutated pan-cancer, while some were tumor-type specific, like FOXA1 in prostate and breast cancer

Biological & Clinical Impact

Mutations in significantly mutated regions correlate with marked changes in expression and downstream signaling, and differential survival and therapeutic outcomes.

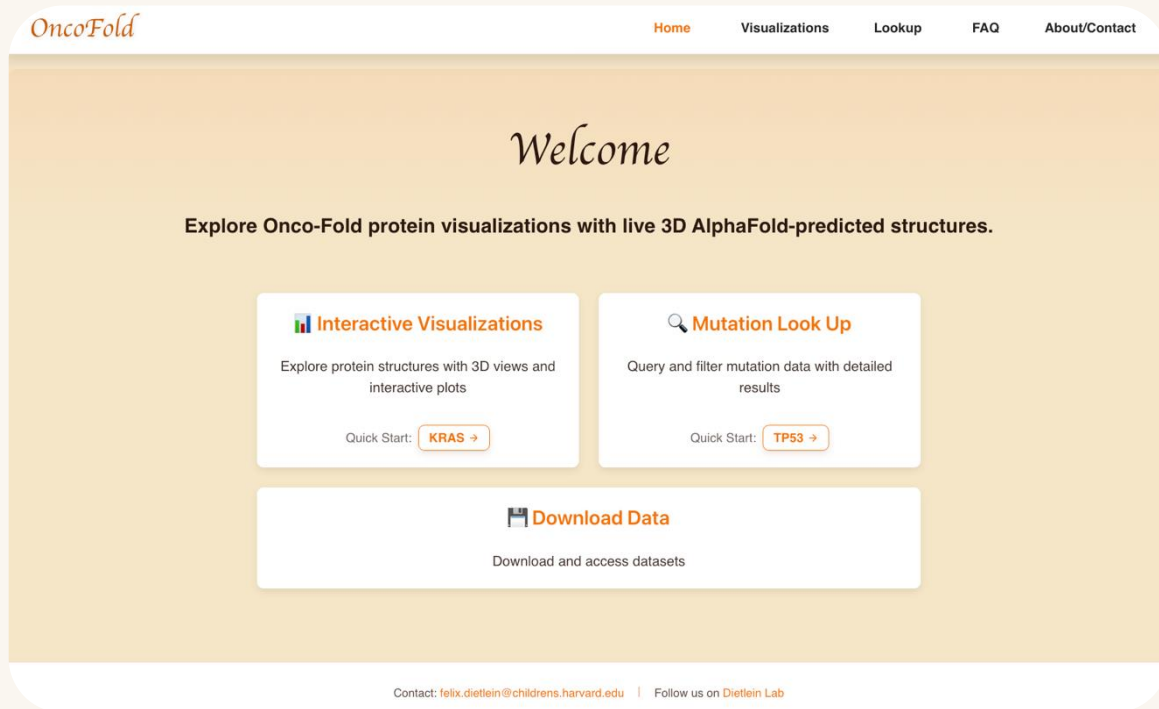


oncofold.org

Interactive Exploration of Somatic Mutations in 3D

Four core features: Visualizations • Lookup • Interact • Download

OncoFold at a Glance



Interactive Visualizations

Multi-panel protein plots with pathogenicity scores, mutation counts, state probabilities, and 3D visualization

Mutation Lookup

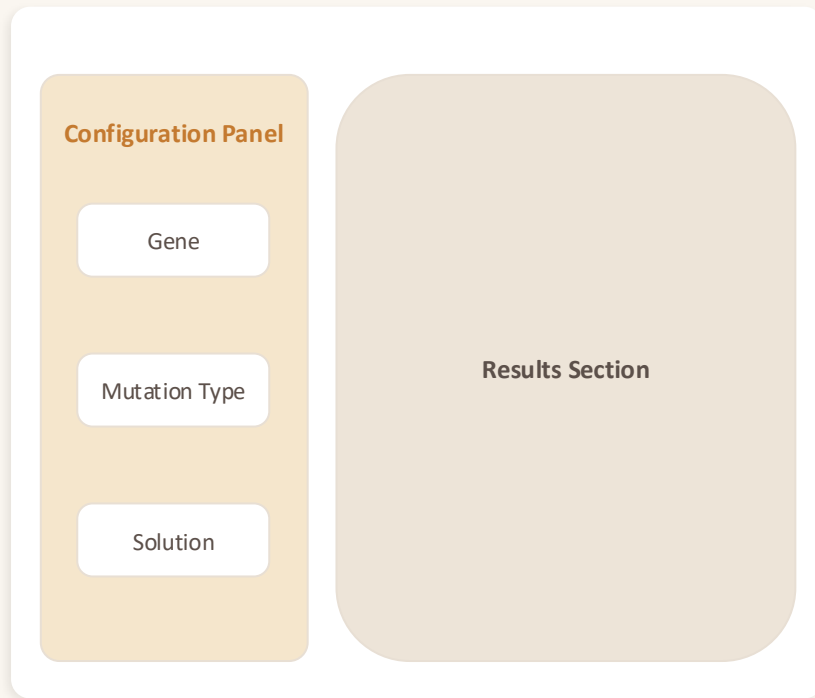
Query mutations by gene, position, amino acid, pathogenicity threshold, and count. Filter for significant regions.

Download Data

Access the results of our analysis for genes with detailed positions and information.

Page Layout Overview

Both the Visualizations and Lookup pages share a consistent two-panel layout.



Visualizations: Getting Started

Quick Start

- 1 Navigate to the Visualizations tab
- 2 Select a gene (e.g., KRAS)
- 3 Plots and 3D structure load automatically
- 4 Adjust mutation type, solution, and display options as needed

Tip: Use Quick Start links on the homepage (e.g., KRAS →) for instant access

Configuration Panel

Gene Selection: Search 173 cancer genes. Icons show structure availability.

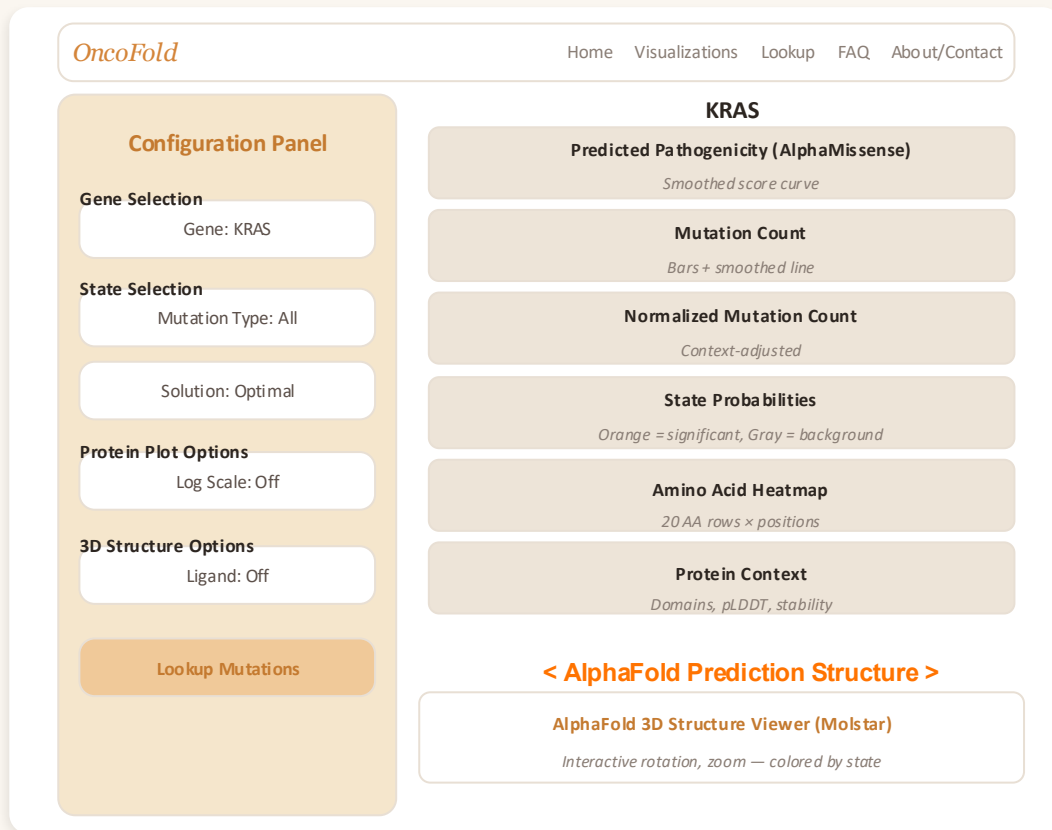
Mutation Type: All mutations, Missense only, or Truncating only.

Solution: Optimal Solution (best fit) or explore alternatives.

Plot Options: Log Scale toggle to reveal subtle mutation patterns.

3D Structure: Toggle With Ligand for bound molecules (e.g., GDP).

Protein Plots: Mutation Landscape



All panels share the same x-axis (protein position) and are interactively linked.

Protein Plots: Deep Dive Panels

Amino Acid Heatmap

Beyond position-level summary:

- Shows exactly which amino acid substitutions at each position drive functional classification
- Rows = 20 amino acids, Columns = protein positions
- Color intensity = mutation frequency
- Identifies specific driver substitutions (e.g., G12V, G12D in KRAS)

Protein Context

Biological context aligned with mutation data:

- Domain annotations
- Secondary structure elements
- Predicted Local Distance Difference Test (pLDDT) confidence scores
- Stability predictions
- Solvent accessibility

Reveals what biological features significantly mutated regions are tied to.

3D Structure Viewer

From linear sequence to spatial insight

< AlphaFold Prediction Structure >



Powered by Molstar

- Rotate, zoom, explore structures in real time
- Orange = significant, gray = background
- Toggle With Ligand for bound molecules
- Supports AlphaFold + experimental (PDB) structures
- Plot selections highlight corresponding 3D regions

Case Study: TP53

TP53 harbors significantly mutated regions in both its DNA-binding domain. In 3D, mutations cluster within the interior core rather than the DNA-binding surface, indicating they destabilize domain structure rather than alter DNA-binding specificity.

Mutation Lookup

From question to proteo-genomic interpretation in seconds

Mutation Lookup

Query mutations by position, amino acid, pathogenicity, or frequency.

Gene & State
Select Gene
TP53
Mutation Type
All mutations
Solution
Optimal S...

Position Range
From
100
To
288
Valid range: 2 - 393

Amino Acid Filter
Reference
Any
Alternative
Any

Thresholds
Min Alpha
Min Count
☒ Significant regions only (non-state0)

Results

Found 957 mutations across 150 positions

Position 103

Ref: Y

→*	7,579,378	α:	1.00	count: 18	state0
→*	7,579,378	α:	1.00	count: 24	state0

Position 104

Ref: Q

→*	7,579,377	α:	1.00	count: 111	state0
→L	7,579,376	α:	0.10	count: 0	state0
→K	7,579,377	α:	0.09	count: 0	state0
→H	7,579,375	α:	0.07	count: 0	state0
→H	7,579,375	α:	0.07	count: 2	state0
→E	7,579,377	α:	0.07	count: 0	state0
→R	7,579,376	α:	0.07	count: 0	state0
→P	7,579,376	α:	0.06	count: 0	state0

Position 105

Ref: G

→R	7,579,374	α:	1.00	count: 12	state0
→C	7,579,374	α:	0.99	count: 48	state0
→V	7,579,373	α:	0.99	count: 68	state0
→D	7,579,373	α:	0.98	count: 17	state0
→S	7,579,374	α:	0.96	count: 17	state0
→A	7,579,373	α:	0.91	count: 10	state0

Position 106

Ref: S

→R	7,579,371	α:	0.15	count: 0	state0
→R	7,579,369	α:	0.15	count: 36	state0
→R	7,579,369	α:	0.15	count: 0	state0
→I	7,579,370	α:	0.10	count: 1	state0
→C	7,579,371	α:	0.07	count: 0	state0
→G	7,579,371	α:	0.07	count: 1	state0
→T	7,579,370	α:	0.07	count: 0	state0
→N	7,579,370	α:	0.06	count: 1	state0

Query Parameters

Gene & State: Select gene + mutation type/solution

Position Range: Narrow to a domain (e.g., 100–288)

Amino Acid Filter: Reference and/or alternative amino acid

Thresholds: Min AlphaMissense score, min count

Significant Only: Filter for significantly mutated positions

Result Cards

Each result displays amino acid change (ref → alt), genomic position, AlphaMissense pathogenicity score, mutation count in the cohort, and assigned functional state.

Download & Resources

Download Data

Complete dataset of significantly mutated regions, mutation rates, and biological variables for all 173 cancer genes.

[oncofold.org](#) → Download

Source Code

Framework with precompiled version, user manual, and test dataset for reproducing key results.

Will be publicly available later

FAQ & About

Common questions about the tool, icons, ligand definitions, amino acid symbols, and contact information.

[oncofold.org](#) → FAQ / About

Live Demo (oncofold.org)

1

Explore Visualizations

2

Compare missense vs. truncating patterns

3

Discover spatial convergence in 3D

4

Look up domain mutations

5

Explore your own gene of interest

Key Takeaways

1

Mutations in cancer genes follow biologically meaningful spatial distributions — not random scatter

2

3D protein structures reveal complex mutational imprints invisible in linear sequence

3

OncoFold provides interactive visualization, lookup, and data download for 173 cancer genes

4

Designed for clinicians, biologists, and computational researchers in precision medicine

Thank You

Q & A

oncofold.org

DoYoung.Hyeon@childrens.harvard.edu

Yuxiang.Zhou@childrens.harvard.edu

Felix.Dietlein@childrens.harvard.edu